

PA1

Mini Discord Network Service

Overview

In this assignment, you will build a simplified Discord-like messaging service called Mini Discord.

Your service must support:

- A custom TCP socket interface.
- An HTTP web-service interface.
- Multiple clients connected at the same time.
- Users and text channels.
- Message creation and message history.
- Detection and handling of failed TCP clients.

Both interfaces must access the same users, channels, memberships, and message history.

You will submit your source code and a standardized recorded demonstration. The demonstration will allow the instructor to grade the required behavior without installing and running every submission.

Scenario

You have been hired to build the backend for Mini Discord, a lightweight communication platform for students in a distributed-systems class.

Users connect to the Mini Discord server and communicate in text channels. Some users interact through a command-line client using TCP sockets. Other users interact through HTTP requests using curl or an automated client.

For this assignment, Mini Discord consists of one server process. A later assignment may extend Mini Discord to multiple servers with server-to-server communication, replication, and distributed failure detection.

System Components

Your submission must include:

- A server with a TCP socket interface.
- A server with an HTTP interface.
- A command-line TCP client

